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Q1: What is Study of trace fossil is called? 2

The study of trace fossils, often called ichnology (from the Greek ichnos, a trace).

Q2: What is meant by Biogeography? 2

Biogeography is the study of the distribution of species and ecosystems in geographic space and through geological time.

Q3: Redundant species hypothesis? 3

According to the *redundant species hypothesis*, the removal of certain species from an ecosystem would have little or no effect on the functioning of that ecosystem. Here, the function of an ecosystem could refer to its overall productivity, its rate of nutrient cycling or its general self‐sustainability.

Q4: What are characteristic features of Pterosaurs? 3

The pterosaurs were proficient flapping flyers, with a lightweight body, narrow hatchet-shaped skull and a long narrow wing supported on a spectacularly elongated fourth finger of the hand. Pterosaurs were covered with hair, and were almost certainly **endothermic**. Most pterosaurs fed on fishes caught in coastal seas, but others were insectivorous.

Q5: Functions of Geographic Information System.

Modern biogeography often employs the use of Geographic Information Systems (GIS), to understand the factors affecting organism distribution, and to predict future trends in organism distribution. Often mathematical models and GIS are employed to solve ecological problems that have a spatial aspect to them.

Q6: History of old world tropical mammal fauna of Africa 5

Africa contains the best known and most distinctive of the Old World tropical mammal fauna. Molecular studies of the interrelationships of the different orders of placentals suggest that the common ancestor of a group of African placentals entered the continent in the Late Cretaceous. This group is made up of the elephants, hyracoids (conies), aquatic sirenians (sea cows), elephant shrews, aardvarks, Cape golden moles, insectivorous tenrecs and extinct embrithopods; they have been placed together in a super order, Afrotheria, that is endemic to Africa.
The early horse *Hipparion*, which had evolved in North America, appeared in both Eurasia and Africa, while rhinoceros, hyenas and sabre‐toothed cats dispersed from Africa to Eurasia.
The elephants originated in Africa in the Eocene, migrated into Eurasia in the Early Miocene and migrated back later in the Miocene. The ancestors of the mammoths and modern Asian elephant probably evolved in Africa 4 to 5 mya and migrated back into Eurasia 2 million years later.

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Q7: What are three types of radiated amniotes 5

Amniotes radiated during the Late Carboniferous, giving rise to three main clades. These are distinguished by the pattern of openings in the side of the skull, especially the **temporal openings** behind the eye socket. These three skull patterns diagnose the key clades among amniotes which are

i. The anapsid (“no arch”) is the primitive state skull pattern, since there are no temporal openings. The Anapsida include various early forms such as *Hylonomus*, as well as some Permian and Triassic reptiles, and the turtles. The two other skull patterns seen in amniotes are

ii. The synapsid (“same arch”), where there is a lower temporal opening. The Synapsida include the “mammal-like reptiles” and the mammals.

iii. The diapsid (“two arch”) pattern, where there are two temporal openings. The Diapsida includes a number of early groups, as well as the lizards and snakes and the crocodiles, pterosaurs, dinosaurs and birds.

Q8: Work of Leon Croizat in field of Biogeography.

He amassed a vast array of distributional data, representing each biogeographical pattern as a line, or track, connecting its known areas of distribution. Croizat studied the distribution patterns of many unrelated taxa, and for each he drew lines or ‘tracks’ on the map linking the areas in which they are found. In many cases, these lines were similar enough in position to be combined as ‘generalized tracks’, shown here. He found that the tracks of many taxa, belonging to a wide variety of organisms, could be combined to form a generalized track that connected different regions of the world

Q9: Write a note on geographical history of New Caledonia

New Caledonia

Geological studies show that New Caledonia is the smallest (about 1600 km2) surviving fragment of Gondwana, having rifted away from Australia in the Late Cretaceous 80–65 mya and reaching its present position 1500 km east of it 50 mya. It is a hotspot for diversity: 77% of its flowering plant species are endemic to the island, as are 98% of the species of the flowering plant family Sapotaceae.

In view of the large amount of research that shows the frequency of long- distance dispersal in such scenarios, most biogeographers today believe that this was the mechanism of arrival of the ancestors of the **New Caledonian taxa.**

Geological studies show that the island was submerged for long periods in the **Paleocene–Eocene**, so that the island was not available for colonization until about 37 mya. The New Caledonian species must therefore have diverged from their ancestors after this date.

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Panbiogeographers, however, **reject long- distance dispersal**. They therefore have to theorize that the Cretaceous ancestors of the present biota were widespread across the Pacific islands, including on supposedly formerly exposed large plateaux such as the Ontong Java plateau.

Biogeographers in Sweden and France have investigated the history of the New Caledonian species of the angiosperm family Sapotaceae. They found that these resulted from nine separate colonizations of the island between 33 and 4.2 mya, dates that are about 40 million years after the rifting of New Caledonia from Australia. New Guinea and Australia were the most important sources of the arrivals, those from Pacific islands arriving only more recently, 27–24 mya. These results are clearly totally incompatible with the panbiogeographers' theories.

Q10: What are two key defining characters of Jawless fishes? 2

Two key defining characters: The head and neural crest tissues.

The neural crest appears in the early embryo as a strip of cells lying just below the outer skin, the ectoderm, of the embryo, above the line of where the backbone will develop.

Q11: Oldest clear evidence for bipedalism

The oldest clear evidence for bipedalism was the find of some human tracks in volcanic ash from Tanzania, dated at 3.75 Ma. The oldest substantial skeletons, of *Praeanthropus afarensis*, come from rocks dated at about 3.2 Ma and also show clear anatomic evidence for advanced bipedalism, but still an ape-sized brain.

Q12: What is Kleptoparasitisim with example? 5

Kleptoparasitism (literally, parasitism by theft) is a form of feeding in which one animal takes prey or other food from another that has caught, collected, or otherwise prepared the food, including stored food (as in the case of cuckoo bees, which lay their eggs on the pollen masses made by other bees; food resources could also be in the form of hosts of parasitic or parasitoid wasps). The term is also used to describe the stealing of nest material or other inanimate objects from one animal by another.
One very widespread duck in both continents is the gadwall. It feeds mainly on submerged vegetation, which is most abundant in nutrient - rich (eutrophic) waters. It cannot dive, so it feeds by upending, and this generally restricts it to the shallow parts of a lake, where it may encounter some competition from other water birds, such as the mallard. Coots (both Eurasian coot and the American coot dive for their food and bring vegetable matter to the surface from greater depths. The coots are messy eaters, and it is not difficult for gadwall to move in and collect some of the loot. This behavior is called **kleptoparasitism**, and it is an effective way of widening the niche of the gadwall.

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Q13: Biogeography of South America 10

South America

The history of the biota of South America has been dominated by the effects of the great earth‐engine of plate tectonics. Some of these effects were caused by the westward drift of the continent, which led to mountain building and consequent climatic changes, but most were the result of changes in its relationship with North America. As noted in the description of the origin of the Caribbean islands in the 'The West Indies' section, there had been islands in that region ever since the Americas started to move westward in the Early Cretaceous. This may have been the route by which some North American dinosaurs dispersed southward in the Mid‐Cretaceous, accompanied by some early types of marsupial and placental mammal. But this connection has always been tenuous, sometimes permitting passage and sometimes becoming broken. This led to cycles of immigration, isolation and evolution, and final new immigration and extinction.

The Early Cenozoic:

The mammal fauna of South America today is characterized by a few marsupials (opossums) and a diversity of edentate placental mammals known as the Xenarthra (sloths, armadillos and South American anteaters) that are hardly known elsewhere. However, in the Late Cretaceous, South America had a mammal fauna of strange marsupials and of ungulate placentals quite unlike anything known elsewhere in the world. Alfred Wallace, one of the founders of zoogeography noted that the range boundaries of Amazonian birds seemed to coincide with the pattern of the rivers. It is fascinating that recent research shows that closely related taxa of some birds are to be found on either side of rivers throughout Amazonia.

Q14: Describe the work of James Dana, and Edward Forbes in the Biogeography of Ocean 5

American scientist **James Dana**, who later became an eminent geologist. His brief paper, published in 1853, divided the surface waters of the globe into several different zones based on their mean minimum temperature.

Three years later 1856, the British zoologist **Edward Forbes** published the first comprehensive work, recognizing five depth zones and 25 faunal provinces along the coasts of the continents. He was the first to recognize the enormous Indo‐Pacific faunal region.

Q15: What is meant by Niche? 3

The demands that an organism places on its environment in terms of physical and chemical conditions, space and food supply help to define what ecologists call its **niche**. But the concept of the niche goes beyond the basic physics and chemistry of its habitat and covers all aspects of how the organism makes a living. It includes the food an animal requires, but also encompasses the way in which it acquires that food. The kestrel is a bird that hunts small mammals by day, whereas an owl performs a similar activity by night. Swallows catch aerial insects by day, and bats have the same feeding strategy at night.

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Q16: What was the brain capacity of Homo habilis? 3

The first species, *H. habilis,* lived in Africa from 2.4 to 1.5 Ma, and had a brain capacity of 630–700 cm3 in a body only 1.3 m tall. *H. habilis* may have used tools.

Q17: Name the most famous fossils. 3

One of the most famous fossils is *Archaeopteryx*, the oldest known bird. The first specimen was found in Upper Jurassic sediments in southern Germany in 1861, and was hailed as the ideal "missing link" or proof of evolution in action. Here was an animal with a beak, wings and feathers, so it was clearly a bird, but it still had a reptilian bony tail, claws on the hand, and teeth. Since 1861, nine more skeletons have come to light, the last two in 1992 and 2005.

Q18: Name the old amniotes and where it was found?

The oldest-known amniote, *Hylonomus* from the mid-Carboniferous of Canada is known only from its skeleton; no amniotic eggs are known from the Carboniferous.

Q19: What is the Biotic Replacement, give example?

Biotic Replacements are an obvious feature of the history of life. These are times when one group of plants or animals replaces another. The replacement of brachiopods by bivalves is a famous example. This had always been seen as a progressive process: the common view is that brachiopods are less adaptable than bivalves, and they clearly succumbed to long-term competition, perhaps lasting for tens or hundreds of millions of years.

Q20: Three amniotic layers in embryo and their functions.

- Enclose the embryo (the amnion)
- Collect waste (the allantois)
- line the eggshell (the chorion)
- Yolk, a yellow material rich in protein

Q21: Trace fossils excrement names, how they are different to other? 3

***Fecal pellets and fecal strings*:** small pellets, usually less than 10 mm in length, or strings of excrement.
***Coprolites*:** discrete fecal masses, usually more than 10 mm in length, and usually the product of vertebrates.

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Q22: Describe the Island biogeography of New Candolle 3

Another early contributor was Candolle, who pointed out that the age, climate and degree of isolation of an island, and whether or not it was volcanic, would also affect the diversity of its flora. Nevertheless, the sheer variety and volume of the works on island biogeography published by Alfred Wallace mark him as the real founder of studies on this subject.

Q23: The work of Albert Gunther, and Arbold Ortman in Biogeography 3

In 1880, the British zoologist **Albert Gunther** published a book on fishes in which he recognized 10 different regions in the distribution of shore fishes.
The German **Arnold Ortmann** published a similar work based on the distribution of crustaceans such as crabs and lobsters.

Q24: When modern birds appear?

Modern groups of birds appeared in the latest Cretaceous and Early Tertiary, including flightless ratites and ancestors of water birds, penguins and birds of prey. The perching birds or songbirds, consisting of 5000 species today, radiated in the Miocene

Q25: Erythrosachus

Small and large meat eaters such as *Erythrosuchus* appeared, one of the first of the archosaurs, a group that was later to include the dinosaurs, pterosaurs, crocodilians and birds.

Q26: Traces on bedding planes

- **Tracks**: sets of discrete footprints, usually formed by arthropods or vertebrates
- **Trails:** continuous traces, usually formed by the whole body, either trave